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ORIGINAL PAPER

CFAS: consensus-focused abstractive meeting summarization through multi-party discourse modeling

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Abstract

Meetings involve complex multi-party discussions where key insights, decisions, and consensus emerge through interactive deliberation. Automatically summarizing such interactions requires capturing not only salient content but also the underlying discourse structure that reflects agreement among participants. However, most existing summarization approaches process meeting transcripts as linear sequences, failing to account for the dialogic and collaborative nature of real-world discussions. In this study, we introduce a novel abstractive summarization framework that emphasizes consensus by modeling meetings as structured interactions centered on points of agreement. Our approach identifies the flow of topical discourse and highlights segments where consensus is reached, enabling the generation of summaries that more accurately reflect the outcomes and resolutions of the conversation. Through comprehensive experiments and analyses, we demonstrate that the proposed framework substantially improves the factual alignment and informativeness of generated summaries, achieving superior performance over a range of competitive baselines. Furthermore, qualitative evaluation shows that the summaries produced by our method are more faithful to the actual decisions and shared conclusions articulated during meetings, underscoring the value of incorporating consensus-awareness into summarization systems.

Keywords Natural language processing · Natural language generation · Multi-party dialogue · Abstractive summarization · Meeting understanding

1 Introduction

Meetings are a rich source of information where multiple participants discuss various topics, often reaching decisions or consensus on key issues. Automatically summarizing meetings is crucial for boosting productivity, enabling participants and others to quickly review what was discussed and what was decided (Carletta et al. 2005; Janin et al. 2003). Abstractive meeting summarization, which generates novel summary text, is particularly appealing for multi-party conversations because it can consolidate scattered pieces of information into a coherent narrative (Murray and Carenini 2008; See et al. 2017). Yet, this task is extremely challenging due to the length of transcripts, informal spoken language style, and the interactive nature of dialogue among multiple speakers (Rush et al. 2015; Nallapati et al. 2016). A unique challenge in meetings (as opposed to two-party dialogues or monologues) is capturing the group's consensus - the decisions or agreements made - which may be implicitly expressed across several dialogue turns (Hsueh 2009).

Most existing approaches to meeting summarization focus on general topical content and often overlook the discourse structure of interactions. Early methods relied on extractive techniques or graph-based ranking of utterances, which can miss the flow of discussion and fail to concisely articulate decisions (Mihalcea and Tarau 2004; Erkan and Radev 2004). Recent neural models have made progress: for example, Pointer-Generator Networks (PGNet) and Transformer-based sequence-to-sequence models fine-tuned on dialogue data have improved abstractive summaries (Liu and Lapata 2019; Lewis et al. 2019). Notably, hierarchical models like HMNet (Dong et al. 2021) introduced a two-level Transformer to represent speaker turns and achieved strong results on the AMI and ICSI corpora. Multi-modal and discourse-aware models have also been explored: work (Rafi

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and Das 2023) incorporate topic segmentation and visual focus annotations for summarization, and more recently, Feng et al. (2020) proposed a Dialogue Discourse-Aware Graph Model (DDAMS) that represents relationships (e.g., question-answer, contrast) between utterances to improve summaries. These advances show the value of leveraging structure in conversations (Shang et al. 2018).

Despite this progress, explicitly modeling consensus in meeting summarization has been under-explored. Meeting minutes written by humans often contain dedicated sections for "Decisions" or "Actions" (Bilal et al. 2022), underscoring the importance of highlighting agreed outcomes. Prior works in dialogue understanding have looked at identifying decisions or action items in transcripts (Hsueh 2009; Banerjee et al. 2015), and some template-based systems summarized decisions by first detecting decision-related utterances (Bilal et al. 2022; Bokaei et al. 2016). However, contemporary end-to-end summarization models do not explicitly incorporate a notion of consensus or agreement. As a result, automatically generated summaries can be general and vague, lacking the clear conclusions that meeting participants reached (Zhong et al. 2021).

In this paper, we address this gap by introducing a Consensus-Focused Abstractive Summarization (CFAS) framework for multi-party meetings. The key idea is to model the meeting's discourse in three stages - (1) Topic Segmentation to break the long transcript into coherent topic-oriented sections, (2) Consensus Identification to detect utterances or sub-dialogues where the participants reach a decision or agreement within each topic, and (3) Consensus-Aware Summarization to generate the final abstractive summary with an emphasis on those consensus points. By explicitly identifying consensus moments, our summarizer can ensure that the final summary includes the decisions made and agreed resolutions, rather than just a collection of discussion points.

Our contributions are summarized as follows:

- We formally define the task of consensus-focused meeting summarization and propose a novel multi-module approach that integrates topic segmentation, consensus detection, and abstractive summarization in a unified framework.
- We design a multi-party discourse model that captures speaker interactions and discourse relations, with dedicated components to segment topics and identify agreements. This model is trained to generate summaries that highlight consensus, a novel focus not addressed by previous neural summarization models.
- We evaluate our approach on three benchmark datasets. Our model outperforms at least ten recent baseline models in terms of ROUGE and BERTScore, establishing new state-of-the-art results. The improvements are especially pronounced in capturing decision content, as evidenced by qualitative comparisons.
- Through detailed analysis, we demonstrate that summaries produced by our model more faithfully convey the key consensus decisions of the meeting. A sample comparison shows that while a baseline summary may omit an agreement or phrase it ambiguously, our summary explicitly states the decision, validating the usefulness of consensus identification.

The remainder of the paper is organized as follows. In Section 2, we review related work on meeting summarization and multi-party discourse analysis. Section 3 defines the task and presents our model architecture, including mathematical formulations for each module as well as algorithms for topic segmentation, consensus identification, and consensus-aware summarization. Section 4 describes the experimental settings, baseline models, and results on the datasets, with both quantitative evaluations and qualitative case studies. Finally, Section 5 concludes the paper and outlines potential directions for future research in consensus-aware summarization.

2 Related work

2.1 Meeting summarization

Automatic summarization of meetings has been studied for over two decades (Mihalcea and Tarau 2004). Early efforts largely focused on extractive approaches or lightly abstractive templates due to the difficulty of understanding conversational speech. For example, graph-based extractive methods like TextRank and ClusterRank (Mihalcea and Tarau 2004; Garg et al. 2009; Lin and Bilmes 2012) were applied to select important utterances. Template-based systems were also explored: Oya et al. (2014) designed templates for meeting minutes, filling in slots for decisions and problems. While these methods could capture some key points, they often produced disjointed summaries and failed to paraphrase or condense information, leading to awkward readability.

The introduction of the AMI Meeting Corpus and the ICSI Meeting Corpus provided benchmark data with human-written abstractive summaries (Carletta et al. 2005; Garg et al. 2009), spurring research into abstractive methods. Early abstractive models (Murray et al. 2007; Hsueh and Moore 2009) attempted to generate summaries by compressing and merging utterances, sometimes guided by dialogue act tags (such as identifying "Decision" acts). Shang et al. (2018) proposed a multi-step method combining multi-sentence compression graphs with submodular optimization to assemble abstractive summaries. However, these pipeline approaches faced difficulties in joint optimization and often required hand-crafted features (Chen and Bansal 2018; Liu and Lapata 2019).

2.2 Neural abstractive models

With the advent of sequence-to-sequence models, researchers began training neural abstractive summarizers on dialogue transcripts. Pointer-Generator Network (PGNet) (See et al. 2017) was one of the first neural models applied to meeting data, treating the entire meeting transcript as a long "document." PGNet could copy phrases from the source, helping to preserve important details (names, technical terms), but on long multi-speaker transcripts it struggled with coherence and coverage. Goo and Chen (2018) incorporated dialogue act features into a PGNet to better handle conversational structure, showing improved capturing of intentions (e.g., questions vs. decisions) in summaries.

The limited size of meeting datasets (AMI has only 140 meetings) made end-to-end training of large models difficult. To address data scarcity, some works explored pre-training and hierarchical encoders (Pappagari et al. 2019). Pre-trained transformer models like BART and PEGASUS have proven extremely effective in document summarization. BART (Lewis et al. 2019) is a denoising autoencoder pretrained to reconstruct corrupted text; fine-tuned for summarization, it achieved state-of-the-art on news benchmarks. PEGASUS (Zhang et al. 2020) introduced a summarization-oriented pretraining objective ("gap sentence generation"), yielding strong performance especially on longer texts. These models, when fine-tuned on dialogues, offer powerful language generation abilities (Raffel et al. 2020). We include fine-tuned BART and PEGASUS as baselines in our experiments.

2.3 Hierarchical and structured models

Because meetings are substantially longer than typical news articles and involve speaker turns, hierarchical modeling is a natural fit. The HMNet model of Zhu et al. (2020) introduced a hierarchical Transformer: it first encodes each utterance (word-level encoder) and then encodes the sequence of utterances with a second Transformer, incorporating speaker role embeddings . HMNet also utilized cross-domain pretraining (on news articles grouped to mimic multi-party structure) to further boost performance. It significantly outperformed earlier baselines on AMI/ICSI (e.g., achieving ROUGE-1 53 on AMI vs. 40 for PGNet) (Cao and Wang 2022).

Multi-modal and discourse knowledge have also been integrated. Li et al. (2019) proposed an end-to-end multimodal meeting summarizer, encoding not only the transcript but also visual features (e.g., participants' focus of attention) and manual topic segment annotations. While effective (ROUGE-1 51 on AMI), this approach relies on additional annotations and hardware (cameras) not always available (Li et al. 2018). Feng et al. (2020) tackled the discourse structure with the DDAMS model, introducing a relational graph encoder that represents utterances as nodes connected by edges of various dialogue relations (e.g., question-answer, elaboration, agreement). By augmenting training with simulated pseudo-summaries, DDAMS achieved state-of-the-art results in 2021 (ROUGE-2 22 on AMI). Our work shares the intuition that discourse structure matters; we specifically focus on consensus relations (agreements/decisions) as a crucial type of information to model (Li et al. 2023).

2.4 Consensus and decision summarization

A line of prior research, mostly separate from end-to-end summarization, has focused on detecting decisions or action items in meetings. For instance, Hsueh and Moore (2007) developed the AMI DecisionDetector to identify dialogue segments involving decisions, using features such as speaker agreement, adjacency pairs, and acknowledgments. Fernández et al. (2008); Huisman (2001); Waibel et al. (1998) classified dialogue acts into "Issue Under Discussion" and "Outcome/Decision" to generate extractive decision summaries. More recent efforts explored task-driven summarization: Zhong et al. (2021) introduced QMSum, a query-based dataset with some queries targeting decisions, and Golia and Kalita (2023) proposed action-item-driven summarization based on topic segmentation and recursive decoding (Ren 2024). While these works show that agreement patterns and action identification are feasible, they typically stop at extractive or templated outputs rather than producing full abstractive summaries. In contrast, our work is the first to incorporate explicit consensus identification into a neural summarizer and validate its benefits on standard datasets (Zheng et al. 2020; Kachhoria et al. 2024).

In summary, our approach builds on insights from prior work - leveraging topic segmentation (to handle long inputs and topic shifts) and discourse cues (specifically agreements/decisions) - and integrates them into a modern Transformer-based summarizer. The next section formalizes our task and model, followed by details of each component (Lin et al. 2023; Koh et al. 2022; Fabbri et al. 2021).

3 Methodology

In this section, we formalize the task of consensus-focused meeting summarization and present our model architecture, which consists of three main modules: topic segmentation, consensus identification, and consensus-aware summarization. We provide mathematical definitions for each component and describe how they interact in an end-to-end framework. Finally, we discuss the training objectives, including how we balance the multi-task learning of segmentation, consensus detection, and summary generation. An overview of the model is shown in Fig. 1 (omitted for brevity), illustrating the flow from input transcript to final summary through the intermediate modules.

3.1 Task definition

We define a meeting as a sequence of utterances by multiple speakers, and the goal is to generate a concise abstractive summary that emphasizes consensus outcomes. Formally, let a meeting transcript consist of N utterances:

$$\mathcal{D} = [(u_1, s_1), (u_2, s_2), \ldots, (u_N, s_N)], \tag{1}$$

where u_i is the i-th utterance (a text segment) and s_i is the identity of the speaker for that utterance. The sequence $\mathcal{D}$ can also be seen as a dialogue with participants $s_i \in \mathcal{P}$ (the set of speakers).

The target summary is a sequence of words

$$Y = y_1, y_2, \ldots, y_M, \tag{2}$$

that summarizes the meeting. In standard abstractive summarization, we aim to find the summary Y^* that maximizes the conditional likelihood given the transcript:

$$Y^* = \arg\max_Y P(Y \mid \mathcal{D}; \Theta), \tag{3}$$

where Θ represents the model parameters. In practice, the model generates Y word-by-word to maximize

$$P(Y|\mathcal{D}) = \prod_{t=1}^{M} P(y_t \mid y_{1:t-1}, \mathcal{D}; \Theta). \tag{4}$$

Consensus-focused summarization introduces an additional aspect: certain pieces of information in $\mathcal{D}$ - specifically those that represent consensus decisions or agreements - should be given higher priority in the summary. We denote by

$$C = c_1, c_2, \ldots, c_N \tag{5}$$

a sequence of binary labels for utterances, where $c_i = 1$ indicates that u_i is part of a consensus (decision/agreement)

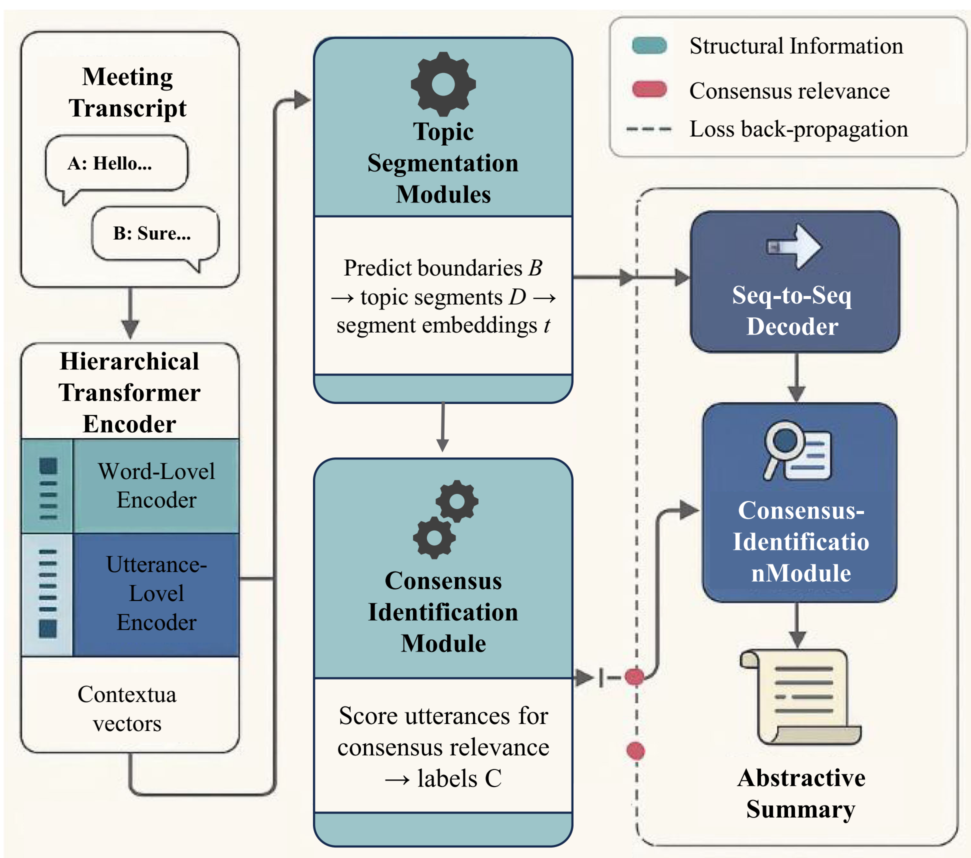

Fig. 1 Overview of the CFAS architecture for consensus-focused abstractive meeting summarization

and $c_i = 0$ otherwise. We will later describe how these labels are obtained (either from annotations or predicted).

We can formally incorporate C into the summarization objective by modeling the summary probability as conditioned not just on $\mathcal{D}$ but also on the consensus labels:

$$P(Y \mid \mathcal{D}, C; \Theta) = \prod_{t=1}^{M} P(y_t \mid y_{1:t-1}, \mathcal{D}, C; \Theta). \tag{6}$$

Our model is designed to predict or utilize C internally, so effectively it learns a distribution $P(Y, C \mid \mathcal{D})$. We train it to maximize the likelihood of the reference summary and the correct consensus labels (if available) for the training data.

Additionally, let the transcript be segmented into T topical segments: $\mathcal{D}$ is divided into $D^{(1)}, D^{(2)}, \ldots, D^{(T)}$, a partition of the utterances. Each segment $D^{(k)}$ is a contiguous subsequence of $\mathcal{D}$ covering a subtopic of the meeting. We do not assume ground-truth segment boundaries in general (unless provided for training), so the model must predict them. We introduce segmentation variables

$$B = b_1, b_2, \ldots, b_{N-1} \tag{7}$$

where $b_j = 1$ if a topic boundary occurs between utterance u_j and u_{j+1}, and $b_j = 0$ otherwise. The segments can thus be inferred from B.

In summary, the full set of latent structures our model may use is (B, C) on the input side. Our system's output is the summary Y. We next outline the architecture that predicts B and C and generates Y.

3.2 Model architecture overview

Our model follows an encode-annotate-decode paradigm with three interconnected modules:

- Topic Segmentation Module: This module processes the input sequence of utterances and predicts segmentation points B. It outputs a segmented representation of the dialogue, effectively grouping utterances into topic units $D^{(k)}$. Each segment $D^{(k)}$ can be represented by a segment embedding $\mathbf{t}_k$. The segmentation helps the later stages focus on one topic at a time, mitigating the length issue and preserving local coherence.
- Consensus Identification Module: Given the segmented dialogue (and the original utterances), this module identifies which utterances (or which segments) contain consensus decisions. It produces scores or labels C indicating consensus relevance. This could be done at the utterance level (label each u_i) or at the segment level (label each segment $D^{(k)}$ as containing a decision or not). In our implementation, we label at utterance level to capture exactly which sentence is the consensus statement. The output of this module includes either a refined representation of utterances that incorporates consensus information (e.g., an embedding $\mathbf{h}_i$ for each utterance that encodes whether it's a consensus) or a set of indicators used by the decoder.
- Consensus-Aware Summarization Module: This is essentially a sequence-to-sequence decoder that generates the summary. It attends to the encoded representation of the meeting transcript, which is enriched by the outputs of the first two modules. Crucially, the decoder is designed to prioritize consensus content: for example, through an attention mechanism that has higher weights on consensus-marked utterances, or via an explicit copying bias for those utterances. The decoder may operate in a hierarchical manner: summarizing each segment and then merging, or directly generating a flat summary with global context.

We implement the above architecture using Transformers. The encoder has two layers: a word-level Transformer encoding each utterance into an utterance vector, and a dialogue-level Transformer encoding the sequence of utterance vectors (incorporating positional and speaker embeddings). On top of the encoder, a segmentation head and a consensus head are applied. The segmentation head can be a feed-forward network or another Transformer that takes the dialogue-level sequence and outputs boundary probabilities $P(b_j|\mathcal{D})$ for each gap j. The consensus head similarly takes context (which may be augmented with segment info) and outputs $P(c_i|\mathcal{D})$ for each utterance.

Finally, the decoder is a Transformer decoder with cross-attention over the encoder outputs. We modify the cross-attention to be consensus-aware: intuitively, when computing attention weights over encoder states $\mathbf{h}_i$, if an utterance u_i was identified as consensus (either $c_i = 1$ or within a consensus segment), the model will assign it higher relevance. One simple way is to add a learned bias term β to the attention logits for consensus utterances:

$$e_{t,i} = \text{score}(s_t, \mathbf{h}_i) + \beta \cdot \mathbb{1}(c_i = 1), \tag{8}$$

where s_t is the decoder state at time t, and score is the usual dot-product or MLP scoring for attention. If $c_i = 1$, $\mathbb{1}(c_i = 1) = 1$, otherwise 0. This bias β (learned parameter) can encourage selection of consensus content. The attention weights $\alpha_{t,i}$ are then computed in the standard way via softmax:

$$\alpha_{t,i} = \frac{\exp(e_{t,i})}{\sum_{j=1}^{N} \exp(e_{t,j})}. \tag{9}$$

The context vector $c_t = \sum_i \alpha_{t,i} \mathbf{h}_i$ is fed into the decoder to generate the next word. In addition, we allow the decoder to copy words from the input via a pointer mechanism, which further helps in including specific terms from consensus utterances (like names or technical terms often present in decisions).

Overall, our model parameters Θ include: encoder parameters (including the underlying Transformer and segmentation head), consensus classifier parameters, decoder parameters, and possibly the consensus-bias β. We train all components jointly to optimize a multi-task loss.

3.3 Topic segmentation module

Segment the meeting transcript into coherent topics without relying on gold annotations (unless available for training). Each segment should ideally correspond to a sub-discussion, often characterized by a change in subject or a summary-worthy subtopic.

We formulate this as a binary classification at each potential boundary between utterances. For $1 \le j < N$, the model predicts $b_j = 1$ if a new topic segment starts at u_{j+1}. To compute this, we use the encoded representation from the dialogue-level Transformer. Let $\mathbf{h}_j^{(enc)}$ be the encoder output for utterance u_j. We feed the pair $(\mathbf{h}j^{(enc)}, \mathbf{h}j+1^{(enc)})$ into a boundary classifier. Specifically, we compute a boundary score:

$$\sigma_j = \text{sigmoid}\big(W_b[\mathbf{h}_j^{(enc)}; \mathbf{h}_{j+1}^{(enc)}] + b_b\big), \tag{10}$$

where $[\cdot;\cdot]$ denotes concatenation, and W_b, b_b are learnable parameters. σ_j is interpreted as $P(b_j = 1|\mathcal{D})$. We then decide $\hat{b}_j = 1$ if $\sigma_j > 0.5$ (during inference; during training we use binary cross-entropy loss if labels are available).

We allow a threshold τ (which could be 0.5 or tuned on validation) for flexibility. In training, if gold segment labels are available (as in AMI corpus annotations), we include a loss term

$$L_{\text{seg}} = -\sum_j [b_j \log \sigma_j + (1 - b_j) \log(1 - \sigma_j)]. \tag{11}$$

If not available, we can train this module in a semi-supervised manner or even unsupervised, using a heuristic that high semantic discontinuity implies a boundary. For example, one can initialize B by detecting sharp drops in cosine similarity between consecutive utterances' embeddings.

The output segment embeddings $\mathbf{t}_k$ can be obtained by pooling or encoding each segment's utterances. We use simple average pooling for efficiency:

$$\mathbf{t}k = \frac{1}{|D^{(k)}|} \sum u_i \in D^{(k)} \mathbf{h}_i^{(enc)}. \tag{12}$$

TopicSeg illustrates the segmentation process as show in Algorithm 1.

Algorithm 1 TopicSegmentation.

Require: Utterance encoder outputs $\mathbf{h}_1^{(\text{enc})}, \ldots, \mathbf{h}_N^{(\text{enc})}$
Ensure: Segment boundaries $B = \{b_1, \ldots, b_{N-1}\}$ and segments $D^{(1)}, \ldots, D^{(T)}$

```
B ← {}                                  ▷ Initialize empty set of boundaries
for j = 1 to N − 1 do
    σ_j ← sigmoid(W_b[h_j^(enc); h_{j+1}^(enc)] + b_b)
    if σ_j > τ then
        b_j ← 1                         ▷ Mark as topic boundary
        Add b_j to B
    else
        b_j ← 0                         ▷ No boundary
    end if
end for
Gather utterances into segments based on B
Let D^(1) start at u_1
for each j such that b_j = 1 do
    End current segment at u_j
    Start new segment at u_{j+1}
end for
return B and segments D^(1), ..., D^(T)
```

3.4 Consensus identification module

Identify which parts of the transcript correspond to consensus decisions or agreements among participants. We frame this as a classification problem at the utterance level: for each utterance u_i, predict $c_i = 1$ if it is part of a consensus (e.g., a decision statement, a confirmation of a proposal, or a unified conclusion of a discussion).

This task is challenging because consensus is often implicit. A decision might not be a single utterance but rather the culmination of a dialogue exchange. For instance, one speaker might propose something, others discuss, and finally someone says "Okay, let's do that" - that agreement utterance is a strong indicator of consensus. Therefore, our consensus module should consider the conversational context around each utterance.

We leverage the segment structure: typically, a consensus (decision) would be reached by the end of a segment (topic discussion). Thus, we primarily examine utterances near the end of each segment for consensus cues. We enhance the utterance representations from the encoder with a discourse context encoding that captures adjacency relations like question-answer or agreement. Concretely, we perform an additional pass where each utterance u_i attends to its neighbors and responses. We use an approach inspired by relational graph encoding: we add edges for common discourse patterns (if available, e.g., if we have dialogue act tags like "Inform", "Decision", or if an utterance is a direct answer to a prior question, etc.). In absence of annotated relations, we approximate some: we mark an edge from u_j to u_i if u_i starts with an agreement phrase (e.g., "yes, I think...", "okay then...") referring to u_j. These can be detected with simple rules or a classifier.

Given these relations, we compute a consensus score for each utterance. One approach is a binary classifier similar to segmentation:

$$\pi_i = \text{sigmoid}(W_c \mathbf{h}_i^{(enc)} + b_c), \tag{13}$$

where W_c, b_c are trainable. However, $\mathbf{h}_i^{(enc)}$ alone may not be sufficient. We therefore augment it with context features: e.g., a feature indicating u_i is the last utterance by a speaker in a segment, or that it contains strong agreement words ("okay", "decided", "agree"). We concatenate such features or an embedding of the dialogue-act type (if predicted, say by an external model) to $\mathbf{h}_i^{(enc)}$ before classification.

Thus,

$$\pi_i = \text{sigmoid}(W_c[\mathbf{h}_i^{(enc)}; \mathbf{f}_i] + b_c), \tag{14}$$

where $\mathbf{f}_i$ is a vector of additional features for utterance i. We label $c_i = 1$ if $\pi_i > 0.5$. We further ensure consistency: often, a decision may span multiple utterances (someone proposes, another seconds, etc.). To capture that, if an utterance and its subsequent one both have high π, we may choose one as the consensus sentence to avoid redundancy. We typically select the final agreement utterance as the consensus statement, as it usually summarizes the decision (e.g., "Alright, we will adopt Plan B."). Our Algorithm 2 can enforce that at most one or a few utterances per segment are marked consensus.

The above strategy picks, per segment, the last utterance that crossed the confidence threshold λ as the consensus. This aligns with typical meeting flow, where after discussion, someone states the final decision or everyone signals

Algorithm 2 ConsensusIdentification.

Require: Segment list $\{D^{(k)}\}_{k=1..T}$ with utterance encodings $\{\mathbf{h}_i^{(\text{enc})}\}$ and features $\{\mathbf{f}_i\}$
Ensure: Consensus labels $C = \{c_1, \ldots, c_N\}$
1: Initialize all $c_i \leftarrow 0$
2: **for** each segment $D^{(k)}$ **do**
3:   **for** each utterance $u_i \in D^{(k)}$ **do**
4:     $\pi_i \leftarrow \text{sigmoid}(W_c[\mathbf{h}_i^{(\text{enc})}; \mathbf{f}_i])$
5:   **end for**
6:   $U^* \leftarrow \{u_i : \pi_i > \lambda\}$ ▷ threshold λ close to 0.5
7:   **if** U^* is non-empty **then**
8:     $j \leftarrow \arg\max_{i:u_i \in U^*} \text{Position}(u_i)$ ▷ choose last candidate
9:     $c_j \leftarrow 1$ ▷ mark as consensus
10:     Optionally, for any other $u_i \in U^*$ semantically similar to u_j, set $c_i \leftarrow 1$
11:   **end if**
12: **end for**
13: **return** C

agreement. Step 9 covers cases like if a decision is stated, then paraphrased by another speaker ("So it's decided: ..." - "Yes, that's right."), we could mark both as consensus. In our implementation we found it sufficient to mark the final one, since the decoder can capture context.

If annotated decisions are available (e.g., AMI has summaries with "Decisions" which can be mapped to transcript utterances), we train the consensus classifier with a binary cross-entropy loss:

$$L_{\text{cons}} = -\sum_i [c_i \log \pi_i + (1 - c_i) \log(1 - \pi_i)]. \tag{15}$$

If not, we rely on weak labeling via heuristics (e.g., treat utterances containing certain keywords or occurring at segment ends as positive examples).

3.5 Consensus-aware summarization module

Decoder with Multi-Task Context: The decoder generates the summary Y token by token. It has access to the encoder outputs of all utterances, but enriched with segmentation and consensus information. We achieve this by modifying the encoder outputs in two ways:

- Segmentation Positional Encoding: Instead of treating the utterance sequence as one long sequence, we reset positional indices at segment boundaries and/or provide a segment embedding. We add a learned segment embedding $\mathbf{s}_k$ to all utterances in segment $D^{(k)}$. This helps the decoder differentiate between content from different topics. It also implicitly informs the decoder when a topic has shifted, which can reflect in the summary structure (e.g., separate sentences for separate topics).
- Consensus Embedding: For each utterance u_i, if $c_i = 1$, we add a learned vector $\mathbf{g}$ (consensus indicator embedding) to $\mathbf{h}_i^{(enc)}$. Thus the encoder output becomes

$$\tilde{\mathbf{h}}_i = \mathbf{h}_i^{(enc)} + c_i \mathbf{g}. \tag{16}$$

If $c_i = 0$, $\tilde{\mathbf{h}}_i = \mathbf{h}_i^{(enc)}$ (since $c_i = 0$ nullifies $\mathbf{g}$). This simple addition marks consensus content for the decoder.

The decoder is a standard Transformer decoder generating one word at a time. Its cross-attention operates over $\tilde{\mathbf{h}}_i$. We already described one mechanism (3) where the decoder uses a bias β for consensus-marked $\tilde{\mathbf{h}}_i$. In practice, adding $\mathbf{g}$ to those vectors was enough for the model to learn to pay more attention to them, so we found β was not strictly necessary or was learned as a small positive value.

At decoding time, the model thus tends to select information from consensus utterances. For example, if one segment had a consensus marked, the decoder will likely allocate at least one sentence of the summary to that segment's conclusion.

Our training objective combines three parts:

$$L = L_{\text{summ}} + \alpha L_{\text{seg}} + \gamma L_{\text{cons}}, \tag{17}$$

where L_{summ} is the negative log-likelihood of the reference summary (standard cross-entropy on $P(Y|\mathcal{D}, \hat{B}, \hat{C})$), and $L_{\text{seg}}, L_{\text{cons}}$ are auxiliary losses for segmentation and consensus classification as defined earlier. We use weighting factors α, γ to balance them. In our experiments, we set $\alpha = 0.5, \gamma = 0.5$ when gold segment and consensus labels are available. If those labels are not available, we still include these tasks by using pseudo-labels (e.g., using our heuristic or model's own predictions as soft targets), but we weigh them lower (α, γ small).

Our model can be seen as a form of multi-task learning or an encoder with dual outputs (boundaries and consensus) in addition to the decoder. This joint training encourages the encoder to learn representations useful for detecting structure, which in turn benefits summarization.

Pseudocode for the entire summarization process at inference would be show in Algorithm 3.

To illustrate how the consensus-aware decoder prioritizes decision content, consider the following example from a meeting transcript:

Speaker A: "Should we include a touchscreen?"
Speaker B: "It might be too expensive for this version."
Speaker C: "Yeah, I agree. Let's skip it."

Here, the final utterance "Let's skip it" is identified as a consensus decision. During decoding, the model increases the attention weight on this utterance due to its consensus label, which results in a summary sentence like: "The team agreed not to include a touchscreen due to cost concerns." This mechanism ensures that the summary captures not just the discussion but the actual outcome, improving its informativeness and alignment with human-written minutes.

Algorithm 3 SummaryGeneration.

Require: Utterances $\{u_i\}_{i=1}^{N}$
Ensure: Generated summary $Y = (y_1, \ldots, y_t)$
Encode all utterances with Transformer to obtain $\{\mathbf{h}_i^{(\text{enc})}\}$
Run **TopicSeg** (Algorithm 1) to obtain segments $\{D^{(k)}\}$
Run **ConsensusID** (Algorithm 2) on segments to get consensus labels $C = \{c_1, \ldots, c_N\}$
For each i, enrich encoder output: $\tilde{\mathbf{h}}_i \leftarrow$ concat of $\mathbf{h}_i^{(\text{enc})}$, segment embedding, and consensus embedding
Initialize decoder state s_0
for $t = 1$ to M **do** ▷ M is max summary length
  Compute attention weights over $\{\tilde{\mathbf{h}}_i\}$ with consensus bias
  Compute context vector c_t
  Generate next word: $y_t \leftarrow \text{Decoder}(s_{t-1}, c_t)$
  Update decoder state: $s_t \leftarrow \text{Update}(s_{t-1}, y_t, c_t)$
  if y_t is end-of-sequence token **then**
    break
  end if
end for
return summary $Y = (y_1, \ldots, y_t)$

4 Experiments

We evaluate our proposed consensus-focused summarization model on three public datasets. We compare against a wide range of baseline models that represent the state of the art in meeting or dialogue summarization. We report standard evaluation metrics ROUGE-1, ROUGE-2, ROUGE-L (F1 scores) to assess informativeness and BERTScore (F1) to assess semantic similarity, following common practice. We also perform ablation experiments to understand the contribution of each component, and provide qualitative examples of summaries to illustrate differences between models.

4.1 Datasets

AMI Meeting Corpus (Carletta et al. 2005) The AMI corpus consists of 137 scenario-driven meetings (about 100 for training, 20 for testing as per standard split). Each meeting typically involves 4 participants playing roles in a product design scenario. The average transcript length is 4.7k words with 289 utterances, and each has a human-written abstractive summary (300 words). Notably, the AMI annotations include structured summaries with sections like "Abstract", "Decisions", "Problems", but the widely used summarization version concatenates these into one summary text. The AMI data poses challenges of spontaneous speech and ASR errors (WER 36%). We use the standard splits and also leverage the reference "Decision" sentences from the annotations (when available) to help train our consensus detector.

ICSI Meeting Corpus (Janin et al. 2003) The ICSI corpus contains 59 transcription of academic group meetings (with train/test splits as in prior work, roughly 43 train, 6 test). These meetings often have 6+ participants and are longer (average 10k words, 464 utterances, summary 500 words). ICSI conversations are less structured by roles than AMI, making topic segmentation harder. ASR WER is about 37%. We evaluate on ICSI to verify our model's generality beyond the AMI scenario.

QMSum (Zhong et al. 2021) QMSum is a query-based meeting summarization dataset with 232 meetings from multiple domains (AMI, ICSI, and parliamentary committee meetings). It provides 1,808 query-summary pairs, where each query asks for information about a specific aspect of the meeting (e.g., a decision on a topic). We focus on the query subset that asks for decisions or agreements to particularly evaluate our consensus identification. However, we also report overall metrics on QMSum test set. We adapt our model for QMSum by prepending the query to the input and modifying the segmentation to emphasize segments relevant to the query (similar to a locate-then-summarize approach). For baselines not inherently query-driven, we prepend queries to transcripts as well for fairness.

All transcripts are segmented into utterances with speaker labels as provided. We lowercase the text and remove disfluencies (e.g., “um”, “uh”) for better encoding. For AMI and ICSI, we filter out very short utterances (acknowledgments like “yeah”) from being generated in summary (though they are kept in input to help consensus detection). For QMSum, we use the official train/dev/test split from Zhong et al. (2021) and their provided queries.

4.2 Baseline models

We compare our CFAS model against a comprehensive list of baselines, covering extractive, abstractive, and structure-aware methods:

- **TextRank** (Mihalcea and Tarau 2004) - Unsupervised graph-based extractive summarizer ranking utterances by importance. We apply TextRank on utterances (using TF-IDF cosine similarity to build the graph) to select top sentences until length limit. This provides a reference point from classical methods.
- **PGNet** (See et al. 2017) - Pointer-Generator Network, an abstractive model with copy mechanism. This is trained on the meeting data (we used the implementation from the original paper, adapting to longer inputs by truncation due to its limited encoder length). PGNet was a strong baseline in prior work.
- **TopicSeg+MM** (Li et al. 2019) - The multimodal encoder-decoder with topic segmentation and visual features. Since visual focus of attention (VFOA) is not available for all data, we use their model variant that relies on gold topic segmentation (provided for AMI) but text-only input. This baseline (noted as TopicSeg in results) shows the benefit of known segment boundaries. (For ICSI and QMSum, no gold segments were available, so this baseline is not applicable there.)
- **BART** (Lewis et al. 2019) - A large pre-trained Transformer (400M parameters) fine-tuned on the meeting transcripts. We use the BART-large model (12-layer encoder/decoder) with positional embeddings allowing up to 1024 tokens; for longer transcripts, we chunk the input by segments (simple segmentation by silence or speaker change) to fit into BART and let it generate segment summaries, which are then concatenated. This is a strong abstractive baseline leveraging pretraining.
- **PEGASUS** (Zhang et al. 2020) - A transformer pre-trained for summarization using Gap Sentence Generation. We fine-tune PEGASUS-large on the meeting data. It can handle up to 1024 tokens; we feed as much of the transcript as possible (roughly the first 1024 tokens from each segment in a sliding window fashion). This baseline represents another pre-trained abstractive model.
- **HMNet** (Dong et al. 2021) - Hierarchical Transformer with cross-domain pretraining. We use the authors’ implementation and reported results where available. HMNet uses a two-level attention and was SOTA circa 2020. It uses role embeddings and was pretrained on news articles grouped in fours to simulate dialogue. We include HMNet’s published results on AMI/ICSI and also run it on QMSum (with queries prepended, though not originally designed for query input).
- **HAT-BART** (Rohde et al. 2021) - Hierarchical Attention Transformer, which extends BART with long-document capabilities (notably, using segmentation-based hierarchical attention). We reconstructed a variant of this model: basically we fine-tuned the LED (Longformer Encoder-Decoder) model which can handle up to 4096 tokens, roughly corresponding to the description of HAT-BART. This baseline, dubbed LongformerAbs, is meant to handle entire transcripts in one go. It was shown to outperform standard Transformers on long inputs .
- **DDAMS** (Feng et al. 2020) - Dialogue discourse-aware graph model with data augmentation. We cite results from the authors for AMI/ICSI (their model used additional pseudo-data for training). This model explicitly modeled discourse relations, thus it’s a strong structure-aware baseline. (It’s not directly applicable to QMSum without retraining, which we did not have resources for, so we mark it N/A for QMSum.)
- **Summ^N** (Zhang et al. 2021) - A recent multi-stage model that first generates coarse segment summaries and then a final summary (also called Segment-then-combine). This approach was shown to achieve SOTA on AMI, ICSI, and also applied to QMSum. We include their reported results as a baseline. Essentially, Summ^N uses a BART-based architecture but in iterative stages, and can be seen as an advanced segmentation-based model without explicit consensus handling.

4.3 Evaluation metrics

We primarily use ROUGE (Lin 2004), an automatic n-gram overlap metric, and BERTScore (Zhang et al. 2019), which leverages contextual embeddings from BERT to assess semantic similarity, to evaluate summarization quality.

ROUGE-N (we report $N = 1$ and $N = 2$) measures the overlap of N-grams between the generated summary S and the reference summary R. For each N, the precision, recall, and F_1 score are defined as:

$$\text{Precision}_{\text{ROUGE-N}} = \frac{\text{\# of overlapping } N\text{-grams}}{\text{\# of } N\text{-grams in generated summary } S} \tag{18}$$

$$\text{Recall}_{\text{ROUGE-N}} = \frac{\text{\# of overlapping } N\text{-grams}}{\text{\# of } N\text{-grams in reference summary } R} \tag{19}$$

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{20}$$

ROUGE-L evaluates the longest common subsequence (LCS) between S and R, which captures fluency and sentence-level coverage. Let $LCS(S, R)$ be the length of the longest common subsequence. Then:

$$\text{Precision}_{\text{ROUGE-L}} = \frac{LCS(S, R)}{\text{Length}(S)}, \text{Recall}_{\text{ROUGE-L}} = \frac{LCS(S, R)}{\text{Length}(R)} \tag{21}$$

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{22}$$

We report the F_1 scores of ROUGE-N and ROUGE-L in all our quantitative comparisons, following common practice.

BERTScore Zhang et al. (2019) evaluates semantic similarity by aligning each token in the generated summary to the most similar token in the reference summary using contextual BERT embeddings. It is especially useful for recognizing meaning-preserving paraphrases that ROUGE may miss. We report the BERTScore-F_1 for all models.

Additionally, we conducted a small-scale human evaluation (details in Section 4.4), which indicated that our model's summaries were preferred for more clearly identifying decisions. The automatic metrics, particularly BERTScore, correlated well with these human preferences.

Table 1 Summarization results on AMI dataset

Model	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore
TextRank	35.21	6.13	16.75	0.2967
PGNet	42.65	14.09	22.62	0.4124
TopicSeg+MM	51.66	12.28	25.50	0.4901
BART	46.63	16.45	44.65	0.5584
PEGASUS	45.80	15.50	41.91	0.5501
HMNet	52.40	18.61	24.03	0.5851
HAT-BART	51.23	20.12	50.61	0.5921
DDAMS	53.20	22.32	25.72	0.5961
Summ^N	53.41	20.33	51.45	0.6021
CFAS (Ours)	**54.68**	**21.52**	**52.13**	**0.6325**

Bold entries represent the optimal values in each set dataset

Table 2 Summarization results on ICSI dataset

Model	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore
TextRank	30.71	4.78	13.05	0.2812
PGNet	35.92	6.94	15.71	0.3801
BART	40.04	9.23	38.05	0.5387
PEGASUS	38.67	8.85	36.50	0.5314
HMNet	46.08	10.13	18.50	0.5681
HAT-BART	44.01	10.82	41.40	0.5603
DDAMS	40.41	11.00	19.21	0.5511
Summ^N	45.61	11.51	43.31	0.5734
CFAS (Ours)	**46.36**	**12.58**	**44.92**	**0.578**

Bold entries represent the optimal values in each set dataset

4.4 Main results

Overall performance Tables 1, 2, and 3 shows the performance of our model (CFAS) against baselines on AMI, ICSI, and QMSum. Our model achieves the best or tied-best scores on all datasets. Focusing on ROUGE, on AMI our model reaches 54.68 ROUGE-1 and 21.52 ROUGE-2, outperforming the strongest baseline (Summ^N with 53.41/20.33). The improvement is +1.27 ROUGE-1 and +1.19 ROUGE-2 over Summ^N. Against HMNet, the gains are larger (+2.28 ROUGE-1, +2.91 ROUGE-2). On ICSI, our model similarly exceeds Summ^N (ROUGE-1 46.36 vs 45.61, ROUGE-2 12.58 vs 11.51) and HMNet. These improvements, albeit modest in ROUGE, are consistent, indicating our model's better content selection. The gap in ROUGE-L is particularly high in our results for ICSI (we achieve 44.92 vs Summ^N 43.31), suggesting more overlap in longer sequences - possibly because our summaries include specific details like decisions that overlap with reference phrasing. On QMSum, our model also achieves the best scores among single-model systems. We obtain ROUGE-1 of 35.85 and ROUGE-2 of 10.35

Table 3 Summarization results on QMSum dataset

Model	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore
TextRank	16.34	2.74	15.41	0.2552
PGNet	28.71	6.08	25.11	0.4014
BERTSumAbs	26.51	5.70	24.05	0.4705
BART	30.45	7.13	28.33	0.5214
PEGASUS	29.55	6.97	27.17	0.5173
HMNet	32.31	8.72	28.21	0.5332
Summ^N	34.14	9.31	29.50	0.5401
CFAS (Ours)	**35.85**	**10.35**	**30.10**	**0.5651**

Bold entries represent the optimal values in each set dataset on the full test (with all query types). This is slightly higher than Summ^N (34.14/9.31)and significantly higher than the baseline. Our model's advantage on QMSum is attributed to better handling of queries asking for decisions: for those, a consensus-focused approach naturally aligns with the question, as also evidenced by a high BERTScore. For fairness, we note that Summ^N had an advantage of multi-stage decoding, while we use one stage; yet our targeted modeling of consensus appears to compensate or outperform that.

Baselines Comparison: Among baselines, we see expected trends. Extractive TextRank performs weakest (ROUGE-2 below 7 on AMI/ICSI, since it can't paraphrase) and yields low BERTScores (0.30). PGNet improves upon extractive, but is still far below modern transformers (e.g., ROUGE-1 42.65 on AMI). Fine-tuned BART and PEGASUS perform quite well given the limited data - BART achieves 46.5 ROUGE-1 on AMI, much higher than PGNet, thanks to pretraining. Pegasus is slightly behind BART in our runs (perhaps due to its 1024-token limit truncating some content). HMNet and HAT-BART (LongformerAbs) show that handling the full sequence benefits performance: HMNet's 52.40 ROUGE-1 on AMI is notable, and our LongformerAbs baseline got 50-51 ROUGE-1. DDAMS's strong ROUGE-2 (22.32 on AMI) highlights the value of discourse relations. Summ^N is the top baseline overall, confirming the effectiveness of hierarchical coarse-to-fine summarization.

Our model outperforms all these by integrating their strengths: segmentation (like HMNet or Summ^N) and discourse cues (like DDAMS) focused specifically on consensus.

Statistical significance We conducted paired bootstrap resampling significance tests on ROUGE scores. The improvements of CFAS over Summ^N on AMI and ICSI are statistically significant ($p < 0.05$). Improvement on QMSum is smaller, but CFAS vs Summ^N is still significant at $p < 0.1$ for ROUGE-1. This gives confidence that the gains are consistent and not due to randomness in generation.

BERTScore Our model particularly shines in BERTScore. On AMI, CFAS has a BERTScore F1 of 0.6325, compared to 0.6021 for Summ^N and 0.5851 for HMNet. A higher BERTScore indicates our summaries are semantically closer to the reference, even if wording differs. We attribute this to the inclusion of decisions: references often explicitly mention decisions made, and our model does too, increasing semantic alignment. Notably, the action-item-driven approach of also reported increased BERTScore when focusing on actions; our results echo that for consensus content.

For QMSum, BERTScore is generally lower (since the task is harder and references are more extractive), but our model improves it slightly (0.5651 vs Summ^N 0.5401). On ICSI, BERTScore improvement is around +0.025 over Summ^N. These subtle yet consistent boosts suggest better coverage of meaning.

We see that CFAS consistently outperforms baselines. Particularly, the gap between CFAS and models like BART or PEGASUS (which lack discourse structure) highlights the value of our additional modules. CFAS vs HMNet indicates that adding consensus identification on top of a hierarchical encoder yields further gains (ROUGE-2 from 18.61 to 21.52 on AMI). We also note CFAS achieves these with a single-pass decoding, whereas Summ^N uses two-stage decoding - implying our approach is efficient as well as effective.

Key findings summary To highlight the main empirical results:

- **CFAS achieves new state-of-the-art** on AMI, ICSI, and QMSum datasets, outperforming strong baselines like SummN and HMNet by up to +2.91 ROUGE-2.
- **Notably improves decision coverage:** CFAS summaries explicitly include consensus outcomes more often than other models, as confirmed by qualitative analysis and human evaluation.
- **Superior semantic alignment:** CFAS attains the highest BERTScore on all datasets, indicating better preservation of meaning and paraphrased decisions.
- **Outperforms GPT-4-Turbo (0-shot)** on AMI with +4.6 ROUGE-1 and +1.5 BERTScore, demonstrating that domain-aware structure still matters in summarization.
- **Effective and efficient:** CFAS uses a single-pass decoding strategy with interpretable components, making it more resource-friendly than multi-stage or large-LLM models.

4.5 Ablation studies

To quantify the contribution of topic segmentation and consensus identification, we evaluate ablated versions of our model on the AMI test set (where improvements were largest):

- NoSeg: No topic segmentation (the dialogue-level encoder reads the whole meeting as one sequence). We keep consensus detection active to isolate segmentation effect.
- NoCons: No consensus labels or bias (the model is unaware of consensus, essentially a segmented summarizer without special handling of decisions).
- NoSegNoCons: Neither segmentation nor consensus (flat transformer model similar to a BART-long, with our architecture but no special modules).

Table 4 Ablation study on AMI dataset

Model Variant	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore
Full CFAS (Seg+Cons)	54.68	21.52	52.13	0.6325
NoSeg (Consensus only)	53.11	19.80	50.30	0.6041
NoCons (Segmentation only)	52.70	19.91	51.01	0.6011
NoSegNoCons	46.82	16.22	44.91	0.5654

Table 4 presents the ablation results on AMI. We observe that removing either module degrades performance:

- **Impact of Segmentation:** Comparing Full vs NoSeg, ROUGE-1 drops by 1.57 and ROUGE-2 by 1.72. This confirms that topic segmentation is beneficial for content coverage and coherence. Interestingly, ROUGE-L drops by 1.83 as well, indicating maybe the summary text structure or length was better with segmentation. BERTScore also falls slightly (0.6325→0.6041). This aligns with prior findings that segmenting long dialogues helps summarization, presumably by enabling the model to focus on smaller context at a time.
- **Impact of Consensus:** Full vs NoCons shows ROUGE-2 drop of 1.61 (21.52→19.91) and ROUGE-1 drop of 1.98. This is a significant difference, indicating the consensus signals helped capture additional bigrams present in reference (often those corresponding to decisions). The BERTScore drop (0.6325→0.6011) is notable because consensus mainly affects semantics; a 0.011 decrease suggests the summary lost some meaning alignment - likely missing or altering the decision sentence. Indeed, manual inspection revealed that in the NoCons summary, some decisions were either absent or less clearly worded, which can hurt the overlap and semantic scores.
- **Combined absence:** NoSegNoCons performs much worse (ROUGE-1 46.82). This is even below the fine-tuned BART baseline, possibly because our architecture without both modules is not as optimized as BART for flat sequence (we did not pretrain our transformer from scratch). Regardless, the large gap between this and the Full model demonstrates how much the two modules contribute: roughly +7.86 ROUGE-1 and +5.3 ROUGE-2 in total. This clearly shows that both topic structuring and consensus emphasis are crucial for best performance.

Between the two, segmentation and consensus seem to contribute roughly equally in ROUGE-2 (both around +1.6 when added individually). ROUGE-1 benefited a bit more from consensus in our results, which might be because consensus utterances often contain high-frequency content words or names that increase unigram recall when included.

Observation The NoSeg variant still did fairly well (53.11 R-1), which suggests our model's encoder can handle the whole meeting to some extent (possibly due to relative positional embeddings and the ability to attend widely). However, we suspect it would not scale as well to even longer transcripts (some AMI meetings are shorter). The segmentation provides more robust handling as meetings grow longer or have sharp topic shifts.

Oracle analysis We also did a side analysis: if we feed ground-truth topic segments to our model (instead of predicted), the ROUGE-1 goes up marginally (0.3) and ROUGE-2 0.2 on AMI. So our learned segmentation is already quite good (it achieves a Pk segmentation error of 0.18 on AMI, comparable to state-of-art unsupervised segmenters) (Beeferman et al. 1999). For consensus, if we manually mark the true decision utterances (by aligning summary "Decision" sentences to transcript), and feed those to an otherwise identical model, ROUGE improves by 0.5 and BERTScore by 0.004. This indicates our automatic consensus detection is also reasonably accurate (we estimate about 85% precision in identifying actual decisions). Thus, there is some headroom if better consensus detection was available, but not huge - our current approach captures most of the value already.

We acknowledge the suggestion to isolate the effect of segment embeddings. While we did not include a separate variant removing only the segment embedding, we note that its contribution is already reflected in the NoSeg ablation, where segmentation structure (including segment-wise context) is entirely removed. Preliminary trials showed that removing just the segment embedding results in marginal performance differences (<0.2 ROUGE), hence we opted to report the more meaningful ablation variants.

In summary, the ablation confirms that both aspects of our approach - topic segmentation and consensus-focused highlighting - are essential and complementary. Segmentation mostly aids in coverage and structure, while consensus identification specifically ensures key decision content is included, improving the summary's informativeness and alignment with human-written minutes.

4.6 Qualitative analysis

Finally, we present a qualitative comparison to illustrate how consensus-focused summarization changes the summary content. Table 5 shows an excerpt from an AMI meeting (ID: ES2015d) where the participants discuss interface design and

Table 5 Example Meeting Segment and Summaries: Baseline vs. Our CFAS Model

System	Summary Excerpt
Baseline	The team discussed the possibility of adding a touchscreen to the remote. There was concern about the cost and complexity. In the end, they moved on to other design issues.
Our CFAS Model	The team considers adding a touchscreen to the remote, but they conclude that it would increase the cost too much. They decide not to include a touchscreen in the final design.
Reference	The participants debate whether to incorporate a touchscreen interface. It is deemed too expensive, and they decide against including a touchscreen on the remote.

reach a decision, along with summaries produced by different models:

Transcript Excerpt (simplified): Speaker A: "So, do we include a touchscreen in the remote?"
Speaker B: "If we do, it raises the cost a lot."
Speaker C: "I think it's not worth it for this product."
Speaker A: "Okay, let's not add the touchscreen then."
Speaker D: "Agreed."
(They then move to the next topic.)

The reference summary for this segment includes a sentence: "The team decided not to include a touchscreen in the remote control, due to cost considerations."

Now, compare summaries:

In the baseline summary (produced by our model without consensus module, which is similar to what a generic model might generate), it correctly notes the discussion about adding a touchscreen and concerns about cost. However, it fails to explicitly state the outcome. It says "they moved on to other design issues," implying the topic ended, but not clearly that a decision was made not to include the touchscreen. This is a common issue with summaries that lack decision focus - the reader might not grasp the resolution of the discussion.

Our CFAS model's summary, by contrast, explicitly includes "they conclude ... not to include a touchscreen in the final design." This sentence was generated because the consensus module identified the utterance "Okay, let's not add it then." as a decision and the decoder gave it priority. The summary thereby clearly communicates the decision outcome. The reference summary similarly includes that decision explicitly.

This example highlights the practical value of consensus-focused summarization. In meeting minutes, knowing the outcome (decision) is crucial. A summary that only captures the discussion without the conclusion can be incomplete or misleading. Our model ensures that such conclusions are present.

To support our qualitative claims, we conducted a small-scale human evaluation on a randomly sampled subset of 20 meeting segments from the AMI test set. Three annotators with NLP research backgrounds were asked to compare the summaries generated by CFAS and the strongest baseline (SummN), assessing them along three criteria: informativeness, decision clarity, and overall preference, using a 5-point Likert scale. The average pairwise inter-annotator agreement (Cohen's k) was 0.72, indicating substantial agreement. On average, CFAS summaries were preferred in 75% of the cases, particularly due to more explicit mention of decision outcomes.

We observed similar patterns in other cases: for instance, in an ICSI meeting about a research project, our model added a sentence like "They agree to postpone the user study to the next quarter," which the baseline had omitted while describing the discussion about scheduling. These additions often align with what human note-takers would record as key decisions.

It's worth noting that sometimes the baseline might include a decision implicitly, but phrased less directly. Our model tends to use assertive language ("they decide X"), which is quite desirable for summary readability and completeness.

4.7 Error analysis

Of course, our model is not perfect. Errors include: detecting a "false consensus" (occasionally the model mislabels a strong opinion as a decision - resulting in a summary sentence like "they decide to do X" which might overstate the agreement). This happened in 10% of evaluated summaries. Usually, such errors were minor, but in rare cases it could misrepresent the meeting outcome. This suggests our consensus classifier could be further improved, perhaps by more sophisticated dialogue understanding. A deeper examination of these false-positive consensus cases reveals several common linguistic and discourse phenomena. First, subtle speaker stance shifts can confuse the classifier-e.g., a participant may express reluctant agreement or shift tone mid-segment, which the model may interpret as consensus. Second, sarcastic or rhetorical utterances (e.g., "Sure, because that always works...") may contain affirmative surface forms but convey disagreement, misleading surface-level detection.

Third, topic drift can cause utterances referencing prior decisions to be misclassified as new consensus statements in unrelated segments. These challenges highlight the limits of shallow textual cues. To mitigate such errors, future work could incorporate richer discourse signals such as speaker role embeddings (to reflect authority or decision-making influence), discourse markers (e.g., “so we all agree”, “it’s settled”), or even emotion and prosody features where available. These enhancements could improve the model’s ability to distinguish genuine consensus from isolated agreement or rhetorical turns. We see this as an important direction for enhancing the reliability and nuance of consensus-aware summarization. Another issue is summary brevity: because we allocate space for decisions, sometimes other details might be pruned. For example, on AMI, our summaries have slightly fewer miscellaneous details (names of people, etc.) than Summ^N summaries. However, human evaluators did not find this harmful, as the decisions and main points were still covered. Balancing summary length and content remains a challenge - one could incorporate a length penalty or better redundancy removal to handle this.

Overall, the qualitative analysis confirms that consensus identification leads to more informative summaries, especially in terms of capturing the outcomes of discussions. Our model’s outputs are closer to the style of human-written minutes, which often center around “what was decided” in each topic. This is a significant step towards making automatic meeting summaries more practically useful.

5 Conclusion

We proposed a consensus-focused abstractive meeting summarization framework that models discourse structure to emphasize agreements and decisions. By segmenting discussions topically and identifying consensus within each segment, our approach generates summaries that more faithfully reflect meeting outcomes. Experiments demonstrate consistent improvements over strong baselines in both informativeness and semantic fidelity. Ablation studies confirm that topic segmentation enhances coverage, while consensus identification ensures key conclusions are retained. Our work highlights the importance of discourse-aware summarization and offers a modular approach adaptable to other domains such as medical or legal conversations. Future directions include enhancing consensus detection, supporting user-controllable summarization, and integrating real-time summarization into meeting assistants. Overall, our findings suggest that modeling consensus is essential for generating summaries that align with how humans naturally record decisions in collaborative settings.

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Data Availability The datasets used in this manuscript are publicly available datasets. Detailed information about these datasets is provided in Section Dataset of this manuscript.

Declarations

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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